

Examiner reports

Q1.

Part (a)(i) It was apparent that some candidates had not covered this part of the specification and they made no progress. Most who had done so, earned the marks here. However a few wrote $(x + 2)^2$ or even $(x - 2)^2$ and then gave the centre as $(-2, 0)$. Some managed to incorporate the 7 with the y term so wrote the coordinates of the centre as $(2, -7)$.

Part (a)(ii) Most candidates were successful in finding the correct radius. However some forfeited one mark by 'meddling' with their equation and putting 18^2 or $\sqrt{18}$ on the right hand side of the equation.

Part(b) This part was rarely attempted. Even where a correct diagram was drawn, few recognised that the chord would be bisected. Many assumed that the triangle was right-angled at the centre of the circle. Others drew tangents instead of a chord.

Part (c)(i) Many made little progress here. However, it was good to see more able candidates coping well. A few fell at the final line writing $(k - 1)$ instead of $(k + 1)$; a few lost a mark by not introducing ' $= 0$ ' as part of the equation of the circle and simply added ' $= 0$ ' at the end of several lines of working so as to match the printed answer. Many made a slip in squaring $(2k - x)$ and some made gross errors such as writing this as $4k^2 + x^2$ or $4k^2 - x^2$. Others 'simplified' the equation to $(x - 2) + (2k - x) = \sqrt{18}$.

Part (c)(ii) Those candidates who made progress here needed both knowledge and algebraic skills and only a small minority completed this part correctly. However more earned some method marks. Use of the correct condition on the discriminant was required but some just tried to solve the equation using the quadratic formula or used ' > 0 ' instead of ' $= 0$ '. A few attempts at completing the square were seen but most failed to equate the expression to zero.

Part (c)(iii) Many candidates who had made no progress in the rest of the question stated that the line would be a tangent to the circle; however several candidates wrote at length about various transformations and completely missed the point.

Q2.

- (a) (i) It was not uncommon to see the gradient written as $-3/-12$, which was then cancelled down to $-1/4$ or 4 . However, most candidates were successful in finding the correct gradient.
- (ii) Most candidates found a correct equation for the line AB but were unable to rearrange it into the given form with integer coefficients. Far too many mistakes are made when candidates rely on using an equation of the form $y = mx + c$ and they may be better advised to remember alternative forms for the equation of a line.
- (b) It had been expected that solving a pair of simultaneous equations would have been routine at this level. It was, therefore, quite disappointing to see the number of arithmetic errors preventing many from finding $(7, 1)$ as the point of intersection.

Q3.

- (a) Most candidates seemed aware of the need to complete the square, but sign errors were very common when finding r^2 .
- (b) Provided r^2 was given as a positive value, marks were awarded for following through from their answers to part (a).
- (c) Many did not have the stamina to obtain the printed quadratic equation giving the x -coordinates of P and Q . They could usually solve the quadratic equation, but it was very common to see the coordinates of P and Q given as $(2, 0)$ and $(3, 0)$ instead of substituting into the equation $y = x + 4$ to find the correct values of y .

Q4.

This was a straightforward question on coordinate geometry intended to get the candidates off to a confident start and many scored full marks. However many made very fundamental errors.

Part (a) Instead of finding the arithmetic mean of the pair of x -coordinates and the pair of y -coordinates, many candidates found the difference of the coordinates before dividing by 2. A few were able to deduce the coordinates of the mid-point from a diagram and some used the half line vector approach successfully.

Part (b) There were many correct answers but a small number were incorrect and some candidates wrote $\sqrt{52} = 4\sqrt{13}$. Notational errors such as $36 + 16 = \sqrt{52}$ were seen and will not always be condoned. It was surprising how many candidates made no attempt at the distance formula.

Part (c)(i) Although the majority of answers were correct, an alarming number of candidates could not simplify $-6/-4$ correctly. It was also very disappointing to see many using difference in x values/difference in y values in order to find the gradient of a line.

Part (c)(ii) Candidates were expected to use their gradient to determine the equation of the line and many had a complete proof. It was not expected that so many would use the equation $y = mx + c$ at this stage. Many fudged their work to get the printed answer. Some imagined that it was sufficient simply to show the required line had the same gradient as they had evaluated in part (c)(i) or to show that **one** of the points satisfied the given equation. Neither of these on their own scored any marks. Verifying two of the facts was given some credit.

Part (d) Although the majority of candidates found the point of intersection, there were many algebraic and sign errors in the attempts seen – adding equations after equalising coefficients being the most common. Many students did not recognise what was required and made no attempt at it.

Q5.

Part (a) Many candidates wrote the statement $mx - 1 = x^2 - 5x + 3$ followed by the printed answer. An intermediate step such as $x^2 - 5x - mx + 4 = 0$ was expected since the result was a printed answer

Part (b) Very few correct solutions were seen to this part of the question. The condition for equal roots was not clearly understood by many and all sorts of algebraic errors occurred when trying to square $-(5 + m)$. A very common mistake after incorrect squaring was to

write $25 + m^2 = 16$ and then final answers of ± 3 were given for m . An alternative approach was often successful. This involved a realisation that the quadratic must be $(x + 2)^2$ or $(x - 2)^2$ and hence the two possible values of m could be found.

Part (c) It was very rare to see a correct description of the line being a tangent to the curve. This might be a very useful question for future candidates to practice in order to understand the geometrical situations and algebraic relationships when a line intersects, does not intersect or is tangential to a given parabola.



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